Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · FIXED EFFECTS

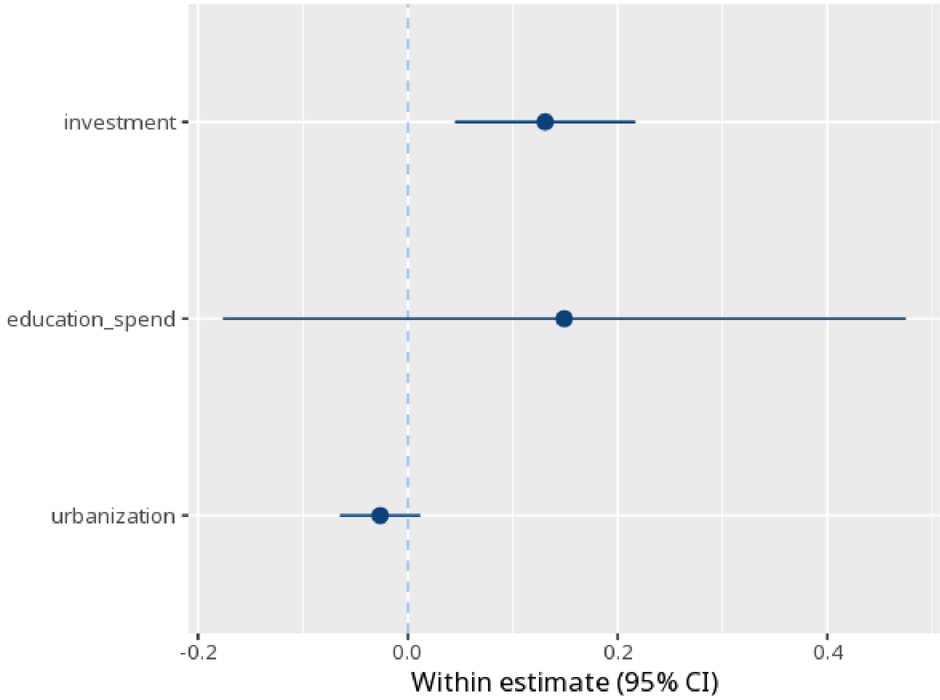
panel regression, entity effects

Table 1. Coefficients (within estimator)

	(1)
investment	0.13
	(0.04)
	[0.04, 0.22]
education_spend	0.15
	(0.16)
	[−0.18, 0.47]
urbanization	−0.03
	(0.02)
	[−0.06, 0.01]
Num.Obs.	100
N entities	10
Within R^2	.44
Adj. within R^2	.36
Between R^2	1.00
Overall R^2	.90
F-test	$F(3, 87) = 22.81$, $p < .001$

time-invariant predictors are absorbed by the entity effects and drop out; a predictor must also change within entities over time to be estimable — predictors with little within-entity variation give large standard errors and unreliable estimates. Standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. F-test for individual effects (poolability): $F = 41.87$, $p < .001$ — a low p favours the entity effects over pooled OLS.

Figure. Coefficients



Understanding the numbers

B. The within-entity effect of a predictor: how the outcome changes when the predictor changes within the same entity over time, controlling for everything stable about that entity. Here, B = 0.13.

Within R^2. The share of the time-varying (within-entity) variation explained by the predictors. Here, within R^2 = .44.

Adj. within R^2. The within R^2, adjusted for the number of predictors - the fairer figure when comparing models with a different predictor count. Here, adjusted within R^2 = .36.

Between R^2. How well the predictors explain differences in the ENTITY-level averages (collapsing out time variation). Here, between R^2 = 1.00.

Overall R^2. How well the predictors explain the outcome using the raw, untransformed data (mixing within- and between-entity variation). Here, overall R^2 = .90.

F-test. The overall F-test of whether the predictors jointly explain more within-entity variation than a model with none. Here, $F(3, 87) = 22.81$, $p < .001$.

N. The total number of entity-period observations used to fit the model. Here, N = 100.

N entities. The number of distinct entities (e.g. firms, countries) whose effects are absorbed. Here, N entities = 10.

Poolability F. A test of whether the entity effects are jointly zero - a low p favours fixed effects over pooled OLS. Here, the poolability F test gave F = 41.87, $p < .001$.

How to read this test

Each B is the within-entity effect of a predictor, controlling for everything stable about each entity. p and the clustered CI assess it; within R^2 is the variance explained by the time-varying predictors. Method: R plm package (Croissant & Millo, 2008). Your significance threshold (α) is 0.05.

APA template: In a fixed-effects model, predictor investment gave B=0.13, p = .003 (clustered SE).

Statistical basis: Fixed-effects (within) panel estimator (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.). Full references in the exported CITATIONS.txt.

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